Template Document

Academic Support Agent

February 19, 2025

1 Experience

April 2017 - Present Machine Learning Engineer Bell

- Focused on developing machine learning models, production deployment, testing, scaling
- Projects: SMS spam detection, Robocall detection

  Tools: Kafka, Flume, Spark, Hive, Solr, Jupyter Lab, Kubernetes, Docker, GitLab CI / CD

  Awards: Merit Gold (3x), Merit Platinum (2x)
- Helped pioneer automated deployments of data pipelines using CI / CD, enabling faster and more portable deployments
- Presented several talks on Machine Learning and Big Data at university hackathons and Bell internal conferences
- Conducted interviews for junior and senior data engineering roles

Sep 2015 - May 2016 Co-founder & Full Stack Developer LocalXChange Inc.

- In a team of 2, raised $8,000 in funding from The Hub incubator at the University of Guelph, to develop a hyperlocal content platform, aimed at delivering local news and events to local users in realtime
- In a team of 3, built hybrid mobile & web apps with Ionic, Angular.js and MongoDB, surpassing 1,000 users within a month since launch
- Met with the Mayor of Guelph and University of Guelph officials to discuss how the app can help boost Guelph Tourism

Sep 2015 - Dec 2016 Graduate Teaching Assistant University of Guelph

- TA for CIS*2430 (OOP), CIS*4150 (Software Reliability & Testing) and CIS*3530 (Database Systems & Concepts) courses
- Delivered several lectures for CIS*3530 when professor was away at a conference

Dec 2013 - Apr 2015 Test Automation Engineer Synechron

- Primarily developed test automation libraries using Java and C#, scripts and CI / CD pipelines
- Lead development of a Keyword Driven & Behavior Driven test framework for Microsoft Dynamics CRM. Earned monetary award & client appreciation.
- Demonstrated that rewriting an in-house test framework for Microsoft Dynamics AX, using an open source library (White) instead of a proprietary one (Coded UI), would help the team save $4K annually by downgrading Microsoft Visual Studio

2 Research

2015 - 2017 MSc. Candidate, Graduate Research Assistant University of Guelph **Thesis**: Data Integration from Multiple Historical Sources to Study Canadian Casualties of WWI

- Proposed a stepwise deterministic method to integrate datasets without labeled data. The method performs comparably with a method that incorporates a Support Vector Machine
- Prepared a longitudinal dataset to enable comprehensive analyses about WWI Canadian society and military, seeding further research
- **Tools**: R, Python, scikit-learn, pandas

3 Publications

L. Antonie, H. Gadgil, G. Grewal, and K. Inwood, Historical Data Integration - A Study of WWI Canadian Soldiers, in 2016 IEEE 16th International Conference on Data Mining Workshops (ICDMW), pp. 186-193, IEEE, 2016.